1 **(a)** $\begin{pmatrix} 8 \\ -6 \end{pmatrix} + \begin{pmatrix} -2 \\ 5 \end{pmatrix}$ **[M1]**

$\begin{pmatrix} 6 \\ -1 \end{pmatrix}$ **[A1]**

(b) $(5, 1)$ **[B1]**

[Total Marks: 3]

2 **(a)** $\begin{pmatrix} 3 \\ 2 \end{pmatrix}$ **[B1]**

(b) $\begin{pmatrix} 6 \\ 4 \end{pmatrix}$ seen or $2\begin{pmatrix} 3 \\ 2 \end{pmatrix} - \begin{pmatrix} -1 \\ 4 \end{pmatrix}$ **[M1]**

$\begin{pmatrix} 7 \\ 0 \end{pmatrix}$ **[A1]**

[Total Marks: 3]

3 **(a)** $\begin{pmatrix} 4 \\ -3 \end{pmatrix} + \begin{pmatrix} -4 \\ 10 \end{pmatrix}$ **[M1]**

$\begin{pmatrix} 0 \\ 7 \end{pmatrix}$ **[A1]**

(b) $\begin{pmatrix} 3 \\ -1 \end{pmatrix} + \begin{pmatrix} 4 \\ -3 \end{pmatrix} = \begin{pmatrix} 7 \\ -4 \end{pmatrix}$ [M1]

$(7, -4)$ **[A1]**

[Total Marks: 4]

4 **(a)** $3 \times \begin{pmatrix} 4 \\ -3 \end{pmatrix}$ **[M1]**

$\begin{pmatrix} 12 \\ -9 \end{pmatrix}$ **[A1]**

(b) $\begin{pmatrix} 4 \\ -3 \end{pmatrix} - \begin{pmatrix} -4 \\ 10 \end{pmatrix}$ or $\begin{pmatrix} 4 - 2(-2) \\ -3 - 2(5) \end{pmatrix}$ **[M1]**

$\begin{pmatrix} 8 \\ -13 \end{pmatrix}$ **[A1]**

(c) $\begin{pmatrix} x \\ y \end{pmatrix} + \begin{pmatrix} -2 \\ 5 \end{pmatrix} = \begin{pmatrix} 6 \\ -1 \end{pmatrix}$ or $(6 - (-2), -1 - 5)$ **[M1]**

$(8, -6)$ **[A1]**

[Total Marks: 6]

5 **(a)** Horizontal component $2 - (-3) = 5$ and vertical component $4 - 1 = 3$ **[M1]**

$\begin{pmatrix}5\\3\end{pmatrix}$ **[A1]**

(b) (i) $2\mathbf{q} = \begin{pmatrix}-2\\8\end{pmatrix}$ **[B1]**

$\begin{pmatrix}1\\6\end{pmatrix}$ **[B1]**

(ii) $3\mathbf{p} = \begin{pmatrix}9\\-6\end{pmatrix}$ **[B1]**

$\begin{pmatrix}10\\-10\end{pmatrix}$ **[B1]**

(c) $\vec{OE} = \begin{pmatrix}2\\4\end{pmatrix} + \begin{pmatrix}-4\\-6\end{pmatrix} = \begin{pmatrix}-2\\-2\end{pmatrix}$ **[M1]**

$(-2, -2)$ **[A1]**

[Total Marks: 8]

6 (a) (i) $\vec{OA} + \frac{1}{2}\vec{AB} = \mathbf{a} + \frac{1}{2}\mathbf{c}$ **[M1]**

$\mathbf{a} + \frac{1}{2}\mathbf{c}$ **[A1]**

(ii) $\vec{MB} + \vec{BN} = \frac{1}{2}\mathbf{c} - \frac{2}{3}\mathbf{a}$ **[M1]**

$\frac{1}{2}\mathbf{c} - \frac{2}{3}\mathbf{a}$ **[A1]**

(b) $\vec{ON} = \vec{OC} + \vec{CN} = \mathbf{c} + \frac{1}{3}\mathbf{a}$ **[M1]**

$\frac{1}{3}$ **[A1]**

[Total Marks: 6]

7 (a) (i) $\mathbf{b} - \mathbf{a}$ **[B1]**

(ii) $\vec{OP} = \vec{OA} + \frac{1}{4}\vec{AB} = \mathbf{a} + \frac{1}{4}(\mathbf{b} - \mathbf{a})$ **[M1]**

$\frac{3}{4}\mathbf{a} + \frac{1}{4}\mathbf{b}$ **[A1]**

(iii) $\vec{MP} = \vec{MO} + \vec{OP} = -\frac{1}{2}\mathbf{a} + \left(\frac{3}{4}\mathbf{a} + \frac{1}{4}\mathbf{b}\right)$ **[M1]**

$\frac{1}{4}\mathbf{a} + \frac{1}{4}\mathbf{b}$ **[A1]**

(b) $\vec{OC} = \vec{OB} + \vec{BC} = \mathbf{b} + k\mathbf{a}$ **[B1]**

$\mathbf{b} + k\mathbf{a} = m\left(\frac{3}{4}\mathbf{a} + \frac{1}{4}\mathbf{b}\right) = \frac{3m}{4}\mathbf{a} + \frac{m}{4}\mathbf{b}$ **[M1]**

Comparing coefficients of $\mathbf{b}$ gives $\frac{m}{4} = 1$, so $m = 4$ **[A1]**

Comparing coefficients of **a** gives $k = \dfrac{3(4)}{4} = 3$ **[A1]**

[Total Marks: 9]

8 (a) $\begin{pmatrix} 8 \\ -1 \end{pmatrix}$ **[B1]**

(b) Deduces correct column vectors for **a** and **b**: $\mathbf{a} = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} -1 \\ 2 \end{pmatrix}$ **[B1]**

Sets up correct simultaneous equations from components: $2p - q = 8$ and $p + 2q = -1$ **[M1]**

Valid method to solve the simultaneous equations (e.g. multiplying first by 2 and adding: $4p - 2q = 16$

5p = 15)$[M1]$

$p = 3$ **[A1]**

$q = -2$ **[A1]**

[Total Marks: 5]

9 (a) Sets up an equation for components, e.g. $2(3) - n = 10$ or $2m - (-2) = 7$ **[M1]**

$n = -4$ **[A1]**

$m = 2.5$ (or $\dfrac{5}{2}$) **[A1]**

(b) Sets up scalar multiplication equation, e.g. $\begin{pmatrix} k \\ 12 \end{pmatrix} = c \begin{pmatrix} 3 \\ m \end{pmatrix}$ **[M1]**

Finds scalar multiplier $c = \dfrac{12}{m}$ **[M1]**

$k = \dfrac{36}{m}$ **[A1]**

[Total Marks: 6]

10 (a) $\begin{pmatrix} 12 \\ -9 \end{pmatrix}$ or $\begin{pmatrix} -4 \\ 10 \end{pmatrix}$ seen **[M1]**

$\begin{pmatrix} 16 \\ -19 \end{pmatrix}$ **[A1]**

(b) $\vec{AB} = \begin{pmatrix} 2 \\ 2 \end{pmatrix}$ **[M1]**

$(5, 1)$ **[A1]**

[Total Marks: 4]

11 (a) $\begin{pmatrix} 12 \\ -9 \end{pmatrix}$ **[B1]**

(b) $\begin{pmatrix} 4 \\ -3 \end{pmatrix} - \begin{pmatrix} -4 \\ 10 \end{pmatrix}$ **[M1]**

$\begin{pmatrix} 8 \\ -13 \end{pmatrix}$ **[A1]**

(c) $(3+4, -1-3)$ **[M1]**

$(7, -4)$ **[A1]**

[Total Marks: 5]

12 (a) $\vec{AB} = \mathbf{b} - \mathbf{a}$ or $\vec{BA} = \mathbf{a} - \mathbf{b}$ **[B1]**

$\vec{OP} = \mathbf{a} + \frac{3}{4}(\mathbf{b} - \mathbf{a})$ or $\vec{OP} = \mathbf{b} + \frac{1}{4}(\mathbf{a} - \mathbf{b})$ **[M1]**

$\frac{1}{4}\mathbf{a} + \frac{3}{4}\mathbf{b}$ **[A1]**

(b) $\vec{MP} = -\frac{1}{2}\mathbf{a} + \vec{OP}$ (oe) **[M1]**

$-\frac{1}{4}\mathbf{a} + \frac{3}{4}\mathbf{b}$ **[A1]**

[Total Marks: 5]

13 (a) $\begin{pmatrix} 6 \\ -4 \end{pmatrix} + \begin{pmatrix} x \\ 5 \end{pmatrix} = \begin{pmatrix} -1 \\ y \end{pmatrix}$ **[M1]**

$6 + x = -1 \implies x = -7$ **[A1]**

$-4 + 5 = y \implies y = 1$ **[A1]**

(b) (i) $(4-2, 7+6)$ **[M1]**

$(2, 13)$ **[A1]**

(ii) $\begin{pmatrix} 2 \\ -6 \end{pmatrix}$ **[B1]**

[Total Marks: 6]

14 (a) Forms a correct equation for the horizontal components: $3(4) - k = 10$ **[M1]**

$k = 2$ **[A1]**

Forms a correct equation for the vertical components: $3(-3) + 2y = 6$

-9 + 2y = 6[$M1$]

$y = 7.5$ **[A1]**

(b) (i) $\begin{pmatrix} 3 \\ 6 \end{pmatrix}$ **[B1]**

(ii) Finds vector $\vec{QR} = \begin{pmatrix} 6 \\ 12 \end{pmatrix}$ OR $\vec{PR} = \begin{pmatrix} 9 \\ 18 \end{pmatrix}$ **[M1]**

States that $\vec{QR} = 2\vec{PQ}$ (or equivalent scalar multiplier), showing the vectors are parallel, AND mentions that they share a common point Q (or P), concluding they are collinear **[A1]**

[Total Marks: 6]

15 **(a)** $\begin{pmatrix} -3 \\ 2 \end{pmatrix}$ **[B1]**

(b) $\vec{BC} = \begin{pmatrix} -9 \\ 6 \end{pmatrix}$ **[M1]**

Identifies method to find C, e.g. $(-1, 5) + \begin{pmatrix} -9 \\ 6 \end{pmatrix}$ **[M1]**

$(-10, 11)$ **[A1]**

(c) $\mathbf{q} = \begin{pmatrix} 4 \\ -1 \end{pmatrix} - \begin{pmatrix} -3 \\ 2 \end{pmatrix}$ **[M1]**

$\begin{pmatrix} 7 \\ -3 \end{pmatrix}$ **[A1]**

[Total Marks: 6]

16 **(a)** $\begin{pmatrix} 7 \\ 10 \end{pmatrix} - \begin{pmatrix} 1 \\ -2 \end{pmatrix}$ **[M1]**

$\begin{pmatrix} 6 \\ 12 \end{pmatrix}$ **[A1]**

(b) $\vec{AC} = \begin{pmatrix} 4-1 \\ 4-(-2) \end{pmatrix} = \begin{pmatrix} 3 \\ 6 \end{pmatrix}$ **[M1]**

$\frac{1}{2}\vec{AB} = \frac{1}{2}\begin{pmatrix} 6 \\ 12 \end{pmatrix} = \begin{pmatrix} 3 \\ 6 \end{pmatrix}$, hence $\vec{AC} = \frac{1}{2}\vec{AB}$ **[A1]**

(c) $(4-5, 4+2)$ **[M1]**

$(-1, 6)$ **[A1]**

(d) $\begin{pmatrix} -1 \\ 6 \end{pmatrix} - \begin{pmatrix} 7 \\ 10 \end{pmatrix}$ or equivalent path method $-\vec{AC} + \vec{CD}$ **[M1]**

$\begin{pmatrix} -8 \\ -4 \end{pmatrix}$ **[A1]**

[Total Marks: 8]

17 **(a)** $\vec{AX} = \frac{1}{3}\mathbf{c}$ **[M1]**

$\mathbf{a} + \frac{1}{3}\mathbf{c}$ **[A1]**

(b) $\vec{BY} = 2\mathbf{a}$ or $\vec{CY} = 3\mathbf{a}$ **[M1]**

$3\mathbf{a} + \mathbf{c}$ **[A1]**

(c) $\vec{OY} = 3\mathbf{a} + \mathbf{c} = 3(\mathbf{a} + \frac{1}{3}\mathbf{c}) = 3\vec{OX}$ **[M1]**

Since $\vec{OY}$ is a scalar multiple of $\vec{OX}$ and they share the common point O, they are collinear. **[A1]**

(d) $1 : 2$ **[B1]**

[Total Marks: 7]

18 (a) $\begin{pmatrix}6\\7\end{pmatrix}$ - $\begin{pmatrix}2\\-1\end{pmatrix}$ $[M1]$

$\begin{pmatrix}4\\8\end{pmatrix}$ **[A1]**

(b) $\vec{AC} = \begin{pmatrix}k-1\\3k-1\end{pmatrix}$ **[B1]**

Gradient or ratio condition: $\frac{3k-1}{k-1} = \frac{8}{4}$ or $\vec{AC} = m\vec{AB}$ **[M1]**

$3k - 1 = 2(k-1) \implies k = -1$ **[A1]**

(c) $\vec{AC} = \begin{pmatrix}-2\\-4\end{pmatrix}$ **[M1]**

$-\frac{1}{2}\vec{AB}$ **[A1]**

[Total Marks: 7]

19 (a) $\vec{PQ} = 2\begin{pmatrix}3\\-2\end{pmatrix} - \begin{pmatrix}-1\\4\end{pmatrix} = \begin{pmatrix}7\\-8\end{pmatrix}$ **[M1]**

$\begin{pmatrix}x\\y\end{pmatrix} - \begin{pmatrix}5\\-1\end{pmatrix} = \begin{pmatrix}7\\-8\end{pmatrix}$ or $(5+7, -1-8)$ **[M1]**

$(12, -9)$ **[A1]**

(b) $\vec{PR} = \begin{pmatrix}14-5\\3-(-1)\end{pmatrix} = \begin{pmatrix}9\\4\end{pmatrix}$ **[B1]**

$3h - k = 9$ and $-2h + 4k = 4$ **[M1]**

Rearranging to get $k = 3h - 9$ and substituting: $-2h + 4(3h - 9) = 4$ or equivalent valid method to solve simultaneous equations **[M1]**

$10h = 40 \implies h = 4$ **[A1]**

$k = 3(4) - 9 = 3$ **[A1]**

[Total Marks: 8]

20 (a) $-\mathbf{a} + \mathbf{b}$ OR $\mathbf{b} - \mathbf{a}$ **[B1]**

(b) $\vec{OD} = \vec{OA} + \vec{AD}$ OR $\vec{OD} = \vec{OB} + \vec{BD}$ **[M1]**

$\vec{AD} = \frac{2}{3}(\mathbf{b} - \mathbf{a})$ OR $\vec{BD} = \frac{1}{3}(\mathbf{a} - \mathbf{b})$ **[M1]**

$\mathbf{a} + \frac{2}{3}(\mathbf{b} - \mathbf{a}) = \frac{1}{3}\mathbf{a} + \frac{2}{3}\mathbf{b}$ **[A1]**

(c) $\vec{CE} = \vec{CO} + \vec{OE}$ OR $-\frac{1}{2}\mathbf{a} + 2\mathbf{b}$ **[M1]**

$-\frac{1}{2}\mathbf{a} + 2\mathbf{b}$ **[A1]**

(d) Finds $\vec{CD} = \vec{OD} - \vec{OC} = (\frac{1}{3}\mathbf{a} + \frac{2}{3}\mathbf{b}) - \frac{1}{2}\mathbf{a} = -\frac{1}{6}\mathbf{a} + \frac{2}{3}\mathbf{b}$ **[M1]**

Shows $\vec{CE} = 3\vec{CD}$ (or equivalent scalar multiple) and states they share a common point C, concluding they form a straight line. **[A1]**

[Total Marks: 8]

21 (a) (i) $\mathbf{a} + \mathbf{b}$ **[B1]**

(ii) $\vec{BC} = \vec{BO} + \vec{OC} = -(\mathbf{a} + \mathbf{b}) + 2\mathbf{b}$ **[M1]**

$\mathbf{b} - \mathbf{a}$ **[A1]**

(iii) $\vec{OM} = \vec{OB} + \frac{1}{2}\vec{BC} = (\mathbf{a} + \mathbf{b}) + \frac{1}{2}(\mathbf{b} - \mathbf{a})$ **[M1]**

$\frac{1}{2}\mathbf{a} + \frac{3}{2}\mathbf{b}$ **[A1]**

(b) $\vec{AC} = \vec{AO} + \vec{OC} = 2\mathbf{b} - \mathbf{a}$ **[B1]**

$\vec{ON} = \vec{OA} + \frac{3}{5}\vec{AC} = \mathbf{a} + \frac{3}{5}(2\mathbf{b} - \mathbf{a})$ **[M1]**

$\frac{2}{5}\mathbf{a} + \frac{6}{5}\mathbf{b}$ **[A1]**

(c) (i) $\vec{ON} = \frac{2}{5}(\mathbf{a} + 3\mathbf{b})$ and $\vec{OM} = \frac{1}{2}(\mathbf{a} + 3\mathbf{b})$, so $\vec{ON} = \frac{4}{5}\vec{OM}$ **[M1]**

$\vec{ON}$ is a scalar multiple of $\vec{OM}$ and both vectors share the common point O, so O, N, and M are collinear **[A1]**

(ii) $\vec{ON} = \frac{4}{5}\vec{OM}$ gives $ON : OM = 4 : 5$ **[M1]**

$4 : 1$ **[A1]**

[Total Marks: 12]